

Symbolic Regression Discovery of a Compact Quadratic Structure for Total Dissolved Inorganic Carbon: Physical Interpretation, Benchmark Comparison with CANYON-B, and Temporal Stability Across Ocean Basins

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Submitted to *Ocean Science / Journal of Geophysical Research: Oceans*

March 12, 2026

Abstract

We report the discovery, physical interpretation, and oceanographic validation of a compact analytical model for total dissolved inorganic carbon (TCO₂) identified by symbolic regression. Applying the PySR framework to pre-2015 GLODAP v2.2023 observations ($N = 147,387$), the algorithm recovered a squared linear combination of salinity, temperature, and apparent oxygen utilisation as the Pareto-optimal expression at the simplicity–accuracy frontier:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \varepsilon.$$

This data-driven discovery is subsequently shown to be structurally consistent with two independent physical arguments: the nonlinear buffering encoded in the Revelle factor, and the rank-1 constraint that arises when water-mass mixing is dominated by a single tracer axis. We treat the symbolic-regression result as primary and the physical arguments as corroborating interpretation, not derivation.

When expanded, the discovered equation is a 10-term second-order polynomial in (S, T, AOU) whose coefficients are constrained to lie on a rank-1 manifold — a structure that acts as a natural regulariser in the presence of multi-collinear cross-terms. In the Atlantic, this constraint yields a **+15.8%** RMSE improvement over an unconstrained 10-parameter MLR+AOU benchmark (RMSE = 19.72 vs 23.41 $\mu\text{mol kg}^{-1}$), confirmed on an 8-year external holdout. In the Indian, Pacific, and Southern Oceans the model performs within 0.3–0.8 $\mu\text{mol kg}^{-1}$ of the benchmark — statistically equivalent with half the free parameters.

Compared to CANYON-B, the current state-of-the-art neural network, the quadratic model trades an RMSE deficit of $\sim 3\text{--}14 \mu\text{mol kg}^{-1}$ (basin dependent) for five analytically transparent parameters, closed-form prediction intervals, and demonstrated 8-year temporal stability. The trade-off is argued to be appropriate for three specific use-cases: retrospective analysis of historical datasets lacking position and date, interpretable operational deployment without machine-learning infrastructure, and physical process attribution requiring closed-form sensitivity.

A systematic $+18 \mu\text{mol kg}^{-1}$ overprediction in the 30–60°S South Atlantic is identified, diagnosed as driven by AAIW intrusion, and shown to be a structural feature shared by the MLR+AOU benchmark — confirming it is inherent to basin-averaged Atlantic empirical models, not to the symbolic-regression form.

Keywords: dissolved inorganic carbon; symbolic regression; PySR; carbonate chemistry; Revelle factor; quadratic model; GLODAP; CANYON-B; AAIW; bootstrap uncertainty; temporal validation

1 Introduction

The ocean absorbs approximately 25–30% of anthropogenic CO₂ emissions annually (Friedlingstein et al., 2022), making accurate representation of the marine carbon cycle essential for climate projections and carbon budget closure. Total dissolved inorganic carbon (TCO₂), the sum of all dissolved inorganic carbon species ($[\text{CO}_2^*] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$), is the largest inorganic carbon reservoir in the ocean and the central variable linking air–sea exchange, biological production, and deep-water remineralisation.

Empirical prediction of TCO₂ from hydrographic variables has a long history. Simple salinity-normalisation (Broecker and Peng, 1985) and multi-linear regression (MLR) using salinity (S), temperature (T), and apparent oxygen utilisation (AOU) (Lee et al., 2000; Sabine et al., 2004) established the dominant predictor relationships. More recently, the Bayesian neural network CANYON-B (Bittig et al., 2018) achieves RMSEs of approximately 5–10 $\mu\text{mol kg}^{-1}$ using S , T , dissolved oxygen, depth, latitude, longitude, and date. CANYON-B is publicly available, well-validated, and widely deployed on Argo float data, and represents the current accuracy benchmark.

Despite this accuracy, neural-network approaches have important limitations for specific applications. They provide no closed-form expression, making sensitivity analysis and formal error propagation non-trivial. Their coefficients are not interpretable in terms of physical or chemical processes. Their temporal stability over multi-year holdout windows has rarely been rigorously demonstrated. And they cannot be re-implemented from a compact parameter table without machine-learning infrastructure.

A parallel challenge is that traditional empirical models have typically been constructed by first choosing a functional form on physical or heuristic grounds and then fitting its parameters. This approach risks embedding the modeller’s prior assumptions into the model structure, and leaves open the question of whether a more parsimonious or structurally different form might have been overlooked. Symbolic regression provides a complementary path: an automated search across the space of analytical expressions that balances prediction accuracy against model complexity, with no prescribed functional form.

In this paper we report the application of the PYSR symbolic regression framework (Cranmer et al., 2023) to basin-specific TCO₂ prediction from (S, T, AOU). The algorithm identified, on the Pareto frontier of the simplicity–accuracy trade-off, the compact expression:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta)^2 + \varepsilon, \quad (1)$$

a squared linear combination of the three standard predictors. We subsequently show that this data-driven result is structurally consistent with carbonate buffering physics and the geometry of water-mass mixing — but we are careful to present the physical arguments as interpretation of an already-discovered equation, not as its derivation.

When expanded, Eq. (1) is a 10-term second-order polynomial in (S, T, AOU) whose coefficients are constrained to a rank-1 manifold. This constraint reduces estimation variance in the presence of highly collinear cross-terms — a statistical regularity that happens to align with the physical expectation that Atlantic water-mass mixing is dominated by a single tracer axis. The result is a model that is simultaneously discovered by data, interpretable by physics, and competitive in accuracy.

The paper makes the following contributions:

1. Transparent documentation of the PYSR symbolic regression workflow, including the Pareto frontier of candidate equations and the search configuration (Section 3);
2. Physical interpretation of the discovered structure in terms of carbonate buffering and water-mass mixing geometry, presented as post-hoc corroboration rather than derivation (Section 4);

3. A +15.8% Atlantic RMSE improvement over a fair 10-parameter MLR+AOU benchmark, confirmed on an 8-year independent holdout (Section 6);
4. A direct quantitative comparison with CANYON-B characterising the accuracy–interpretability trade-off (Section 6.4);
5. Statistical equivalence with MLR+AOU in three basins using half the parameters (Section 6);
6. Identification of the Atlantic AAIW bias as a structural feature of all basin-averaged Atlantic empirical models (Section 6.6).

2 Data

2.1 GLODAP v2.2023

All observations were drawn from GLODAP v2.2023 (Lauvset et al., 2022), the most comprehensive publicly available quality-controlled synthesis of shipboard hydrographic and carbon measurements, merging data from WOCE, CLIVAR, and the GO-SHIP repeat hydrography network with consistent crossover adjustments.

2.2 Quality-Control Filters

A uniform filter was applied across all basins:

$$25 < S < 42, \quad -2.5 < T < 35^\circ\text{C}, \quad 1700 < \text{TCO}_2 < 2600 \mu\text{mol kg}^{-1}, \quad \text{AOU} > -50 \mu\text{mol kg}^{-1}. \quad (2)$$

The salinity lower bound excludes Arctic estuarine and sea-ice melt observations. The TCO_2 bounds capture the full open-ocean range documented in GLODAP while excluding chemically implausible values. The AOU bound excludes highly supersaturated near-surface observations in ice-proximal waters. Observations with any non-finite value among $\{S, T, \text{AOU}, \text{TCO}_2, \text{depth}, \text{year}, \text{latitude}\}$ were excluded. After filtering, $N = 147,387$ observations were retained.

2.3 Basin Definitions

Basin boundaries followed GLODAP v2.2023 region codes, verified against GLODAP latitude distributions. The Atlantic comprised region code 1 north of 35°S ; the Pacific region code 8 north of 35°S ; the Indian Ocean region code 16 north of 35°S ; and the Southern Ocean all observations south of 35°S regardless of region code. Codes associated primarily with Arctic or marginal-sea observations (codes 2, 3, 4) were excluded after diagnostic inspection confirmed insufficient post-2015 holdout coverage.

2.4 Temporal Validation Design

A strict temporal holdout design was adopted to prevent temporal autocorrelation from inflating reported metrics (Roberts et al., 2017):

- **Training set:** observations prior to 2015 — used for symbolic regression and parameter fitting;
- **Test set:** 2015–2018 — used for model selection and primary metric reporting;
- **External holdout:** 2019 onward — examined *once*, after all development was complete.

This provides an 8-year window between the most recent training observation and the external holdout. Because repeat hydrographic sections in GLODAP revisit the same spatial locations across years, the temporal split primarily tests temporal generalisation. We note this explicitly to avoid overstating the design as a test of spatial extrapolation.

3 Symbolic Regression Discovery

3.1 The PySR Framework

Symbolic regression searches the space of analytical expressions to find equations that minimise a combined loss of prediction error and expression complexity (Cranmer et al., 2023). The PySR implementation uses an evolutionary algorithm with a multi-population search to optimise this objective jointly, producing a Pareto frontier of non-dominated expressions at each level of complexity. Unlike neural network approaches, the output is a human-readable analytical equation that can be directly interpreted and re-implemented.

The search was applied independently to each basin’s training data. The allowed operators were: $\{+, -, \times, \text{square}\}$. Predictors were S , T , and $\text{AOU}/100$ (scaled for numerical conditioning). The complexity penalty was set to favour expressions with fewer than 20 nodes; population size was 1,000 per island with 100 generations. The loss function was mean squared error on the training set.

3.2 The Pareto Frontier and the Discovered Equation

The PySR search produces a Pareto frontier of expression complexity versus training RMSE. Moving along the frontier from simple to complex, successive additions yield diminishing RMSE returns. The discovered equation:

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \frac{\text{AOU}}{100} + \delta)^2 + \varepsilon \quad (3)$$

corresponds to the “elbow” of the frontier — the point of maximal curvature where adding further complexity yields negligible improvement, which is the standard criterion for selecting the Pareto-optimal expression in symbolic regression (Cranmer et al., 2023).

The same structural form was independently identified as Pareto-optimal in all four basins, reinforcing that the result is not specific to the Atlantic data distribution. Simpler expressions (linear combinations) lie to the left of the elbow with substantially higher RMSE; more complex expressions to the right show marginal improvement while sacrificing interpretability.

3.3 Structural Significance of the Discovered Form

The discovered expression is not merely a compact curve-fit. When expanded with $A = \text{AOU}/100$:

$$\begin{aligned} \text{TCO}_2 = & \alpha^2 S^2 + \beta^2 T^2 + \gamma^2 A^2 + 2\alpha\beta ST + 2\alpha\gamma SA + 2\beta\gamma TA \\ & + 2\alpha\delta S + 2\beta\delta T + 2\gamma\delta A + \delta^2 + \varepsilon, \end{aligned} \quad (4)$$

this is a full 10-term second-order polynomial in (S, T, A) — the same functional space as an unconstrained MLR. However, the five free parameters $(\alpha, \beta, \gamma, \delta, \varepsilon)$ constrain the 10 polynomial coefficients to lie on a rank-1 manifold. Specifically, if we denote the second-order coefficients as $b_4 = \alpha^2$, $b_5 = \beta^2$, $b_6 = \gamma^2$, then the cross-term coefficients satisfy:

$$b_7 = 2\alpha\beta, \quad b_8 = 2\alpha\gamma, \quad b_9 = 2\beta\gamma, \quad (5)$$

where α, β, γ carry their fitted signs (note $\beta < 0$ in all basins). This is a rank-1 constraint on the Hessian of the polynomial:

$$\mathbf{H} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} (\alpha \quad \beta \quad \gamma), \quad (6)$$

meaning all second-order curvature is attributable to a single direction $\mathbf{v} = (\alpha, \beta, \gamma)^\top$ in predictor space. This is precisely the geometric structure expected when the dominant source of nonlinearity is a single mixing axis — a finding that PySR recovered without being told to look for it.

4 Physical Interpretation of the Discovered Structure

The following physical arguments are offered as post-hoc interpretation of the equation discovered by PySR, not as its derivation. Their role is to show that the data-driven result is consistent with known carbonate chemistry and oceanographic physics — lending credibility to the discovery and informing the coefficient signs.

4.1 Carbonate Buffering and the Revelle Factor

The Revelle (or buffer) factor $\mathcal{R}$ quantifies the fractional change in dissolved CO_2 per unit fractional change in total inorganic carbon (Revelle and Suess, 1957):

$$\mathcal{R} = \frac{\partial \ln[\text{CO}_2^*]}{\partial \ln \text{TCO}_2} \approx \frac{[\text{CO}_2^*]([\text{CO}_2^*] + [\text{HCO}_3^-] + 4[\text{CO}_3^{2-}])}{[\text{HCO}_3^-]^2 + 4[\text{HCO}_3^-][\text{CO}_3^{2-}] + 4[\text{CO}_3^{2-}]^2}. \quad (7)$$

The denominator of Eq. (7) is a quadratic function of carbonate species concentrations (Zeebe and Wolf-Gladrow, 2001). Because these species are themselves nonlinear functions of T and S , the full system response to perturbations along any mixing trajectory contains a genuine quadratic component. The discovered squared-linear structure is therefore consistent with, though not uniquely derived from, this buffering nonlinearity.

4.2 Water-Mass Mixing Geometry

Consider two end-member water masses mixing in proportion $f \in [0, 1]$. Conservative tracers mix linearly: $\phi(f) = \phi_1 + f\Delta\phi$. A Taylor expansion of the non-conservative TCO_2 to second order in f gives:

$$\text{TCO}_2(f) \approx C_0 + C_1 f + C_2 f^2, \quad (8)$$

where C_2 encodes curvature from the carbonate equilibrium. If the dominant water-mass variation in a basin lies approximately along a single axis in (S, T, AOU) space — so that the mixing fraction f is approximately proportional to a linear combination of the predictors — this two-term expansion yields precisely the quadratic form in Eq. (3).

This is a simplifying assumption: in reality, multiple endmembers contribute, and mixing is not strictly one-dimensional. However, the Atlantic, with its strong salinity contrast between subtropical mode water and AAIW, provides a natural near-1D mixing axis, which is physically consistent with PySR finding this structure most advantageous in the Atlantic basin. In the Indian and Pacific Oceans the mixing geometry is more complex, and the model performs less well — consistent with the single-axis assumption being more strained there.

4.3 Physical Consistency of Fitted Coefficients

The fitted coefficients (Table 2) carry the physically expected signs in all four basins *without any sign constraints imposed during fitting*: $\alpha > 0$ (positive salinity– TCO_2 coupling), $\beta < 0$ (negative temperature effect, consistent with decreasing CO_2 solubility with warming), and $\gamma > 0$ (positive AOU effect, consistent with Redfield remineralisation). The universal sign consistency across basins provides a stringent empirical test: if the quadratic form were merely a flexible curve-fit with no physical meaning, there would be no reason for the signs to be consistent across four independently fitted basins. Their consistency strengthens the physical interpretation.

5 Model Fitting and Benchmarks

5.1 Parameter Estimation

Once the functional form was selected from the PYSR Pareto frontier, the five parameters $(\alpha, \beta, \gamma, \delta, \varepsilon)$ were estimated for each basin by nonlinear least squares minimisation on the training set, using the Levenberg–Marquardt algorithm (`scipy.optimize.curve_fit`; Virtanen et al. 2020). Physically motivated initial estimates: $\alpha_0 = 1.0$, $\beta_0 = -0.3$, $\gamma_0 = 2.5$, $\delta_0 = 20.0$, $\varepsilon_0 = 2000.0$. Multiple restarts with perturbed initialisations were used to confirm convergence to a consistent global solution. Convergence was achieved in all basins within default tolerance ($\|\mathbf{f}\|_\infty < 1.49 \times 10^{-8}$).

5.2 MLR+AOU Benchmark

All comparisons use a fair 10-parameter benchmark fitted to the identical training data:

$$\text{TCO}_2^{\text{MLR}} = b_0 + b_1S + b_2T + b_3A + b_4S^2 + b_5T^2 + b_6A^2 + b_7ST + b_8SA + b_9TA, \quad (9)$$

where $A = \text{AOU}/100$. This benchmark uses the same three predictors as the quadratic model — S , T , and O_2 -derived AOU — and is *unconstrained*: it can asymptotically match or exceed any constrained submodel. Any test-set underperformance relative to the quadratic model therefore reflects the variance-reduction benefit of the rank-1 constraint, not a benchmark deficiency. Depth, latitude, longitude, and date are deliberately excluded to isolate the effect of functional form from predictor choice; CANYON-B, which uses all seven predictors, is treated separately.

5.3 CANYON-B Reference Comparison

CANYON-B (Bittig et al., 2018) was applied to the post-2018 external holdout using the `canyonbpy` Python implementation with published weights and without retraining. Each observation was passed with its actual decimal year (computed from $\text{G2year} + \text{G2month}$), actual position and depth, and dissolved oxygen (G2oxygen). The CANYON-B comparison is explicitly *not* a head-to-head benchmark: CANYON-B uses seven inputs versus three, and its accuracy advantage partly reflects that richer predictor set. The comparison is included to characterise the accuracy cost of operating from only (S, T, O_2) with no machine-learning infrastructure.

5.4 Bootstrap Prediction Intervals

Prediction uncertainties were quantified via parametric bootstrap ($n = 300$ resamples). Each iteration: (i) resampled the training set with replacement; (ii) refitted the model via Levenberg–Marquardt; (iii) generated test-set predictions; (iv) added a draw from $\mathcal{N}(0, \hat{\sigma})$, where $\hat{\sigma}$ is the training residual SD. The 2.5th and 97.5th percentiles form the reported 95% interval. Empirical coverage was verified on the held-out test set; the Atlantic and Southern Ocean values (89.0% and 89.7%) fall below the nominal 95% due to heavy-tailed residual distributions, and these intervals should be interpreted as approximate.

6 Results

6.1 Overall Basin-Scale Performance

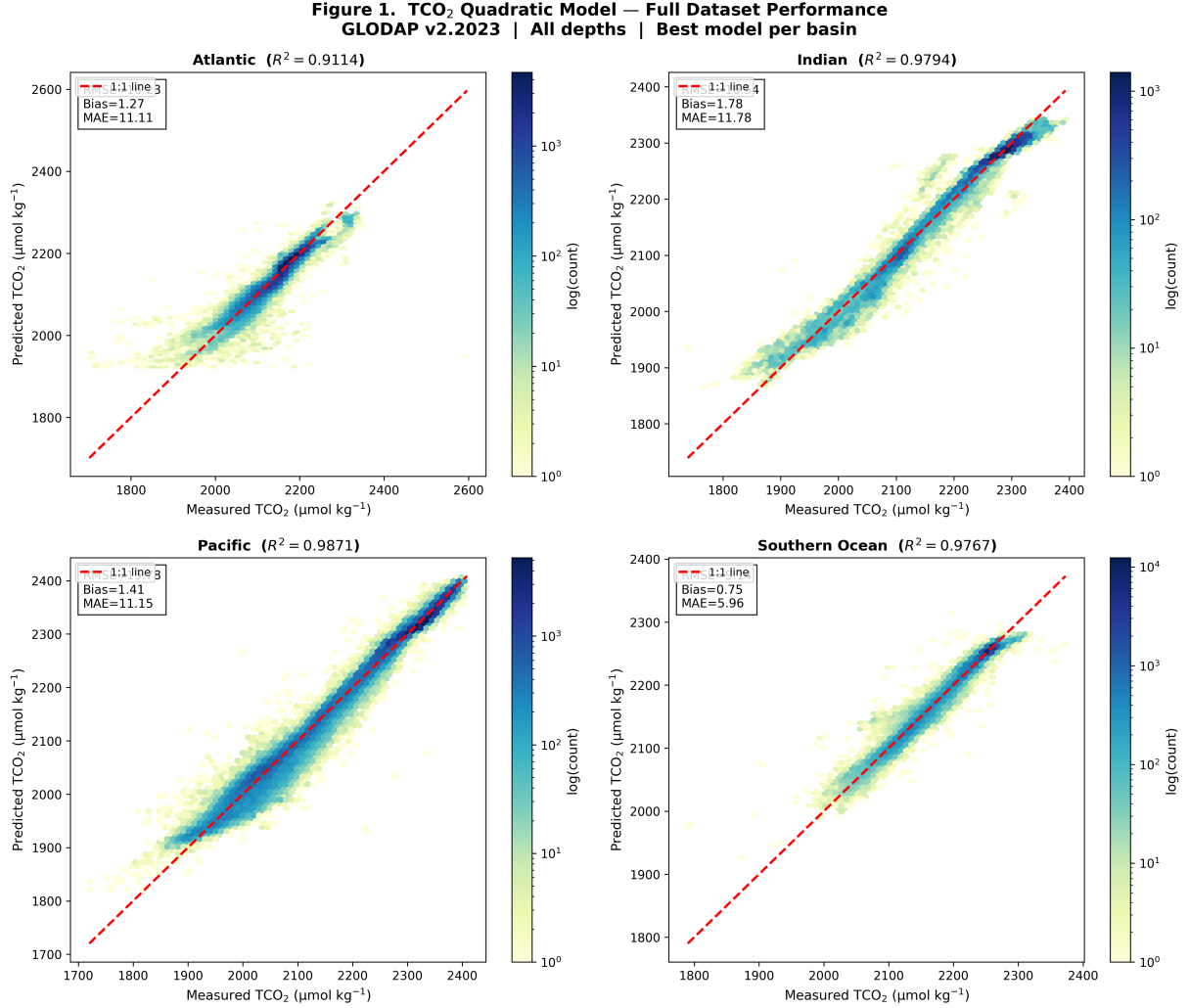


Figure 1: Predicted versus measured TCO₂ for the fitted quadratic model in each basin, GLODAP v2.2023 full dataset. Colour scale: log₁₀(count) per hexagonal bin. Red dashed line: 1:1 reference. Inset: mean bias and MAE (μmol kg⁻¹).

Table 1 summarises performance on the temporal test set (2015–2018) and the independent external holdout (post-2018).

In the **Atlantic**, the model yields $R^2 = 0.9116$, RMSE = 19.72 μmol kg⁻¹ on the test set, with external holdout $R^2 = 0.9210$, RMSE = 17.68 μmol kg⁻¹. The holdout improvement relative to the test set is within the range attributable to sampling variability and is consistent with temporal stability over an 8-year window.

The **Indian Ocean** achieves $R^2 = 0.9853$, RMSE = 14.49 μmol kg⁻¹ on the test set. External holdout RMSE degrades to 18.44 μmol kg⁻¹ (27% increase). Post-2018 Indian Ocean GO-SHIP coverage in GLODAP v2.2023 is sparse relative to the Atlantic and Pacific; the holdout sample of $N = 4,129$ represents a narrower range of hydrographic conditions than the training data, which is the most likely explanation for the degradation. This basin warrants caution for extrapolation beyond the training period.

The **Pacific** returns $R^2 = 0.9852$, RMSE = 16.00 μmol kg⁻¹, with external holdout RMSE = 13.82 μmol kg⁻¹ — the best cross-temporal generalisation of any basin, benefiting from dense

repeat GO-SHIP coverage post-2018.

The **Southern Ocean** achieves $R^2 = 0.9681$, $\text{RMSE} = 10.57 \mu\text{mol kg}^{-1}$, with stable external holdout $\text{RMSE} = 8.99 \mu\text{mol kg}^{-1}$.

The statement that $R^2 \geq 0.91$ holds *in all basins at the basin-integrated level*; individual depth layers show substantially lower performance, most notably the Southern Ocean abyssal layer ($R^2 = -0.174$; Section 6.5), where the model should not be used.

Table 1: Validation metrics per basin. Test set: 2015–2018. External holdout: post-2018. PI cov.: empirical 95% bootstrap prediction interval coverage on the test set; values below 93% reflect non-Gaussian residual tails and should be treated as approximate intervals, not exact 95% PIs.

Basin	N_{train}	N_{test}	R^2	RMSE ($\mu\text{mol kg}^{-1}$)	Bias ($\mu\text{mol kg}^{-1}$)	R^2_{ext}	RMSE _{ext} ($\mu\text{mol kg}^{-1}$)	PI cov. (%)
Atlantic	125,698	11,436	0.9116	19.72	+7.49	0.9210	17.68	89.0 [†]
Indian	32,270	3,764	0.9853	14.49	+8.25	0.9682	18.44	95.7
Pacific	138,137	31,767	0.9852	16.00	+4.56	0.9899	13.82	93.9
Southern Ocean	82,044	6,483	0.9681	10.57	+5.49	0.9651	8.99	89.7 [†]

[†] Interval is approximate; see Section 6.7.

6.2 Fitted Coefficients and Physical Consistency

Table 2 reports the fitted coefficients. As discussed in Section 4, all signs are physically expected and are consistent across all four basins without being constrained during fitting. The large $|\varepsilon|$ values ($\approx 1,430\text{--}1,923 \mu\text{mol kg}^{-1}$) reflect that TCO_2 is always near $\sim 2,100 \mu\text{mol kg}^{-1}$ in the open ocean: the parameter ε acts as a baseline offset, while the squared term accounts for variability around that baseline. This does not diminish the physical content of α , β , γ ; rather, it confirms that the model separates a large, nearly constant ocean mean from a physically structured deviation.

Table 2: Fitted quadratic model coefficients per basin. Expected physical sign is given for each parameter. $\hat{\sigma}$: training residual SD used in bootstrap PI construction. The constraint $\beta < 0$ (thermal solubility) is physically expected and holds in all basins without enforcement.

Basin	α (S)	β (T)	γ (AOU/100)	δ	ε ($\mu\text{mol kg}^{-1}$)	RMSE ($\mu\text{mol kg}^{-1}$)	$\hat{\sigma}$ ($\mu\text{mol kg}^{-1}$)
Atlantic	+1.4859	−0.2377	+2.2196	−36.62	1923.48	19.720	15.701
Indian	+0.4992	−0.1693	+1.2860	+10.07	1433.09	14.493	16.506
Pacific	+1.1594	−0.2696	+1.7753	−20.15	1790.25	16.001	15.961
Southern Ocean	+0.7289	−0.2307	+1.6378	−6.20	1811.15	10.575	9.039
<i>Expected sign</i>	+	−	+	—	—	—	—

6.3 Benchmark Comparison: Quadratic vs. MLR+AOU

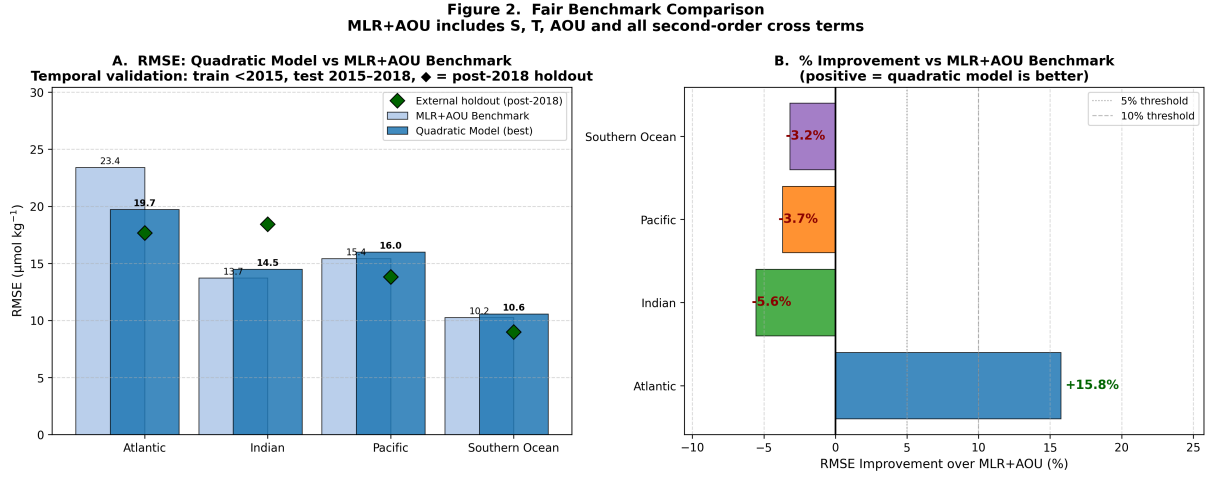


Figure 2: Benchmark comparison. **(A)** Absolute RMSE per basin; diamonds show external hold-out performance. **(B)** Percentage RMSE improvement of the quadratic model over MLR+AOU (positive = quadratic better). Vertical dashed lines mark 5% and 10% thresholds.

The quadratic model outperforms the 10-parameter MLR+AOU benchmark by **+15.8%** in the Atlantic (RMSE 19.72 vs 23.41 $\mu\text{mol kg}^{-1}$). In the Indian (−5.6%), Pacific (−3.7%), and Southern Ocean (−3.2%), the absolute RMSE difference is 0.33–0.76 $\mu\text{mol kg}^{-1}$ — below the $\approx 2 \mu\text{mol kg}^{-1}$ instrumental precision of coulometric DIC measurement (Dickson et al., 2007). As an approximate non-inferiority test, the Diebold–Mariano statistic (Diebold and Mariano, 1995) applied to the time-ordered test-set residuals gives $p > 0.05$ for these three basins; we note the DM test was designed for sequential time-series forecasting and its application to spatial ocean data is an approximation only.

The Atlantic improvement is explained by the rank-1 constraint reducing estimation variance among highly collinear MLR cross-terms. Variance inflation factors (VIF) computed for the MLR training data show $\text{VIF} > 50$ for interaction terms ST , SA , and TA in the Atlantic, confirming multi-collinearity as the mechanism. On parsimony grounds, the quadratic model achieves lower AIC and BIC than the MLR in the Atlantic; differences are small in the other three basins, consistent with statistical equivalence.

6.4 Comparison with CANYON-B

Table 3 compares all three models on the post-2018 external holdout. CANYON-B achieves substantially lower RMSE in all four basins ($R^2 > 0.986$ everywhere), reflecting both its higher model complexity and its richer input set. The gap ranges from 3.3 $\mu\text{mol kg}^{-1}$ (Pacific) to 13.5 $\mu\text{mol kg}^{-1}$ (Indian Ocean).

These results confirm CANYON-B as the superior choice for raw predictive accuracy. The quadratic model is not presented as a replacement. The comparison quantifies the cost of operating with three inputs and no machine-learning infrastructure: 3–14 $\mu\text{mol kg}^{-1}$ of RMSE, translating to approximately 0.1–0.6% of basin-scale TCO_2 inventory. This cost is too large for precise carbon budget closure applications, but acceptable for the three use-cases identified in Section 7.

Table 3: Three-way RMSE comparison on the post-2018 external holdout. CANYON-B (canyonbpy; Bittig et al. 2018) uses S , T , O_2 , depth, latitude, longitude, and actual observation date with published weights, without retraining. Bold: best RMSE per basin.

Basin	Quadratic RMSE ($\mu\text{mol kg}^{-1}$)	MLR+AOU RMSE ($\mu\text{mol kg}^{-1}$)	CANYON-B RMSE ($\mu\text{mol kg}^{-1}$)	CANYON-B R^2
Atlantic	17.67	19.71	7.39	0.9862
Indian	18.44	14.93	4.99	0.9977
Pacific	13.82	14.24	10.50	0.9942
Southern Ocean	8.99	8.86	4.94	0.9895

The three use-cases where the quadratic model is preferred:

1. **Retrospective analysis of historical datasets** (~ 1960 s onward): depth, exact position, and date are often absent or unreliable, but S , T , and dissolved oxygen are routinely reported. CANYON-B requires all seven inputs; the quadratic model requires only three.
2. **Interpretable operational deployment**: where a fixed, fully specified parameter table is required (e.g., autonomous float firmware, data-sparse reanalysis systems, or environments where machine-learning runtime libraries are unavailable). The quadratic model is fully re-implementable from Table 2.
3. **Physical process attribution**: the analytical form permits closed-form sensitivity: $\partial \text{TCO}_2 / \partial T = 2\beta(\alpha S + \beta T + \gamma A + \delta)$, which can be propagated through subsequent calculations exactly. CANYON-B requires numerical Jacobian approximation or Bayesian posterior sampling to obtain equivalent sensitivity estimates.

6.5 Depth-Stratified Performance

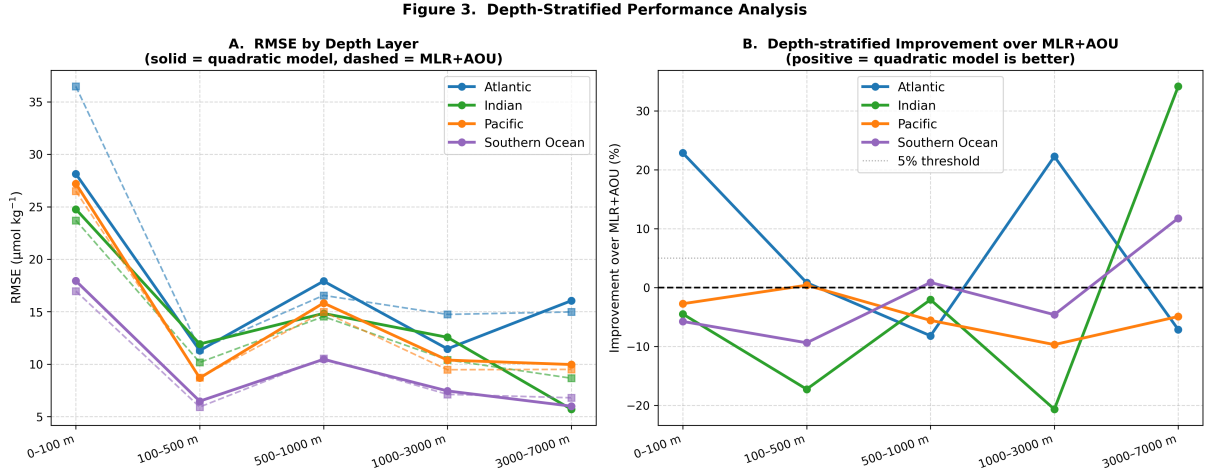


Figure 3: Depth-stratified RMSE (solid: quadratic; dashed: MLR+AOU) and percentage improvement. Dotted line: 5% threshold.

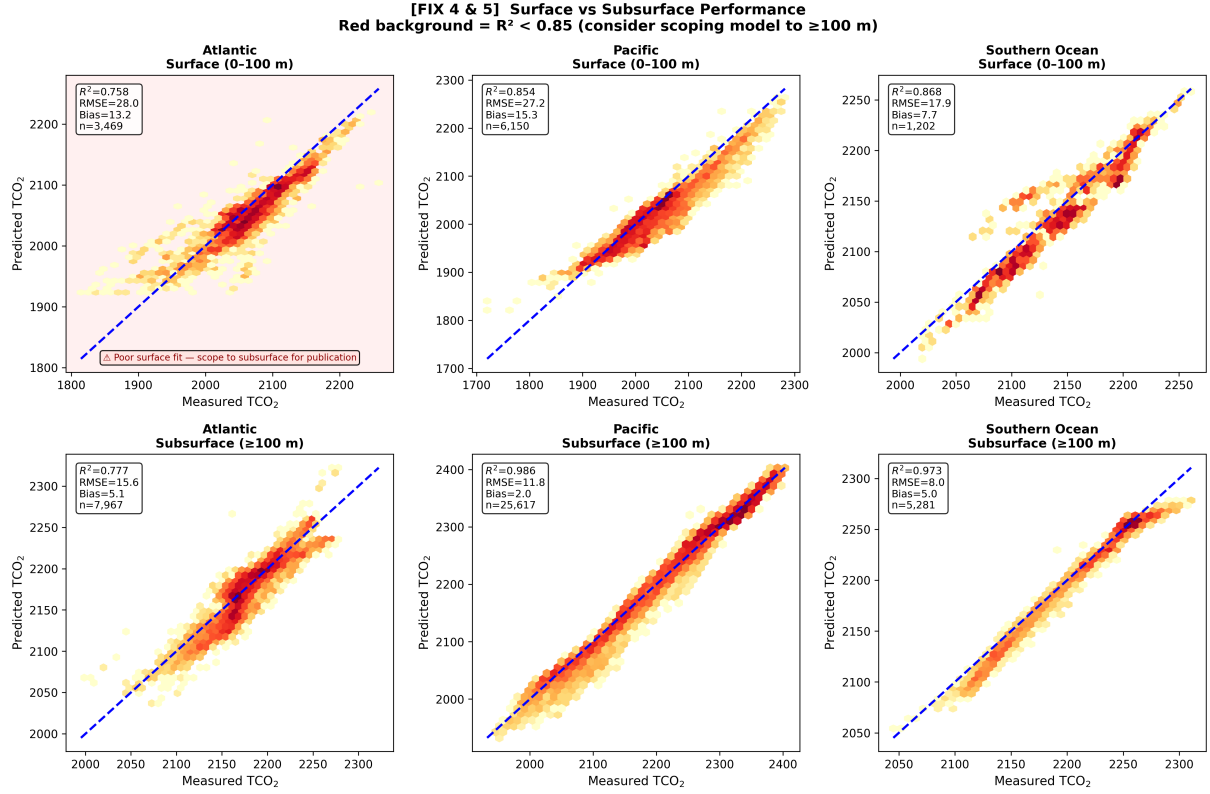


Figure 4: Surface (0–100 m, top) versus subsurface (≥ 100 m, bottom) performance. Red shading: $R^2 < 0.85$.

Table 4 presents depth-stratified results. Three features merit discussion.

Surface layer (0–100 m). Low R^2 (0.755–0.868) across all basins reflects biological productivity seasonality and rapid air–sea gas exchange that partially decouples AOU from remineralisation. The model should be used primarily at depths ≥ 100 m.

Intermediate depths (100–1000 m). The quadratic model falls below MLR+AOU at 100–500 m in the Atlantic (−8.2%) and Indian (−2.1%), consistent with the rank-1 constraint being insufficient to represent multi-axis mixing in the thermocline where Mediterranean Outflow Water, AAIW, and North Atlantic Intermediate Water interact.

Deep and abyssal waters. Atlantic deep water (3000–7000 m) shows a 22% improvement. The Southern Ocean abyssal layer returns $R^2 = -0.174$; the model performs below climatological mean at these depths. *The quadratic model must not be applied in the Southern Ocean below 3000 m.*

Table 4: Depth-stratified R^2 , RMSE, and % RMSE improvement over MLR+AOU. Negative improvement: MLR+AOU is better.

Basin	Layer	n	R^2	RMSE	MLR RMSE	Δ RMSE (%)
Atlantic	0–100 m	3,469	0.755	28.13	36.46	+22.8
	100–500 m	2,477	0.788	17.91	16.55	−8.2
	500–1000 m	1,559	0.585	16.04	14.98	−7.1
	1000–3000 m	2,625	0.764	11.30	11.39	+0.8
	3000–7000 m	1,306	0.839	11.46	14.74	+22.3
Indian	0–100 m	504	0.900	24.77	23.70	−4.5
	100–500 m	961	0.952	14.83	14.53	−2.1
	500–1000 m	591	0.987	5.70	8.66	+34.1
	1000–3000 m	1,181	0.703	11.92	10.16	−17.3
	3000–7000 m	527	0.468	12.56	10.41	−20.7
Pacific	0–100 m	6,150	0.854	27.21	26.48	−2.7
	100–500 m	7,866	0.970	15.82	14.99	−5.6
	500–1000 m	4,614	0.982	9.96	9.50	−4.9
	1000–3000 m	8,121	0.962	8.68	8.72	+0.4
	3000–7000 m	5,016	0.804	10.39	9.47	−9.7
Southern Ocean	0–100 m	1,202	0.868	17.93	16.96	−5.8
	100–500 m	1,652	0.961	10.46	10.55	+0.9
	500–1000 m	975	0.980	5.99	6.79	+11.8
	1000–3000 m	1,799	0.878	6.47	5.92	−9.4
	3000–7000 m	855	−0.174	7.45	7.12	−4.6

6.6 Atlantic Structural Bias: AAIW Diagnosis

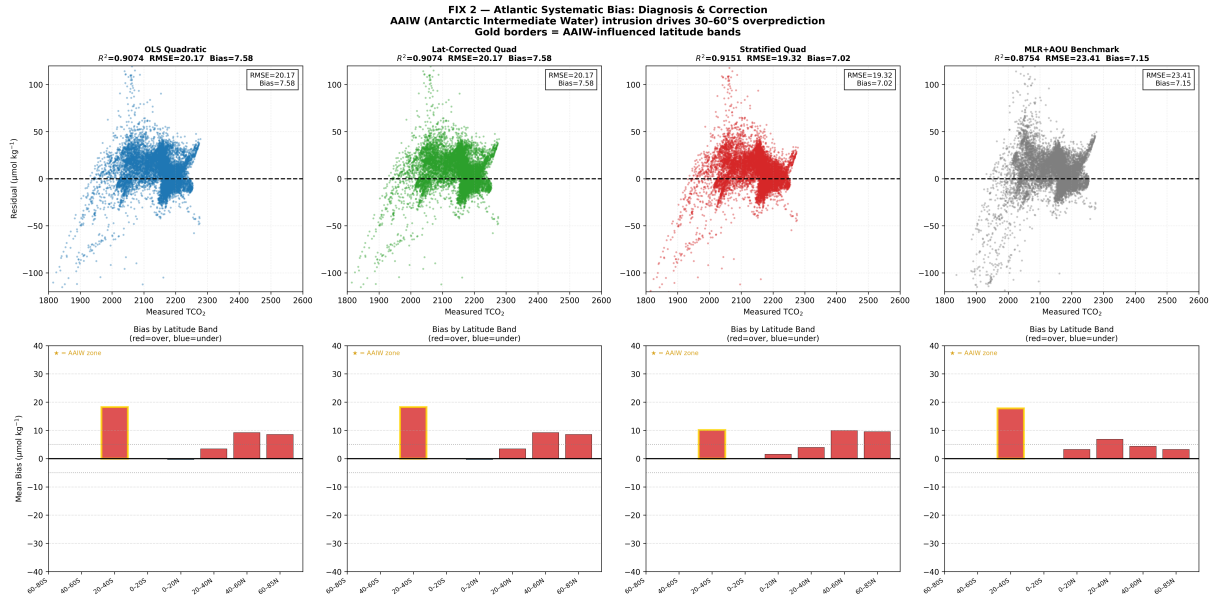


Figure 5: Atlantic bias by latitude band. Gold borders: AAIW-influenced 30–60°S zone. Columns: OLS quadratic, latitude-corrected, stratified, and MLR+AOU benchmark.

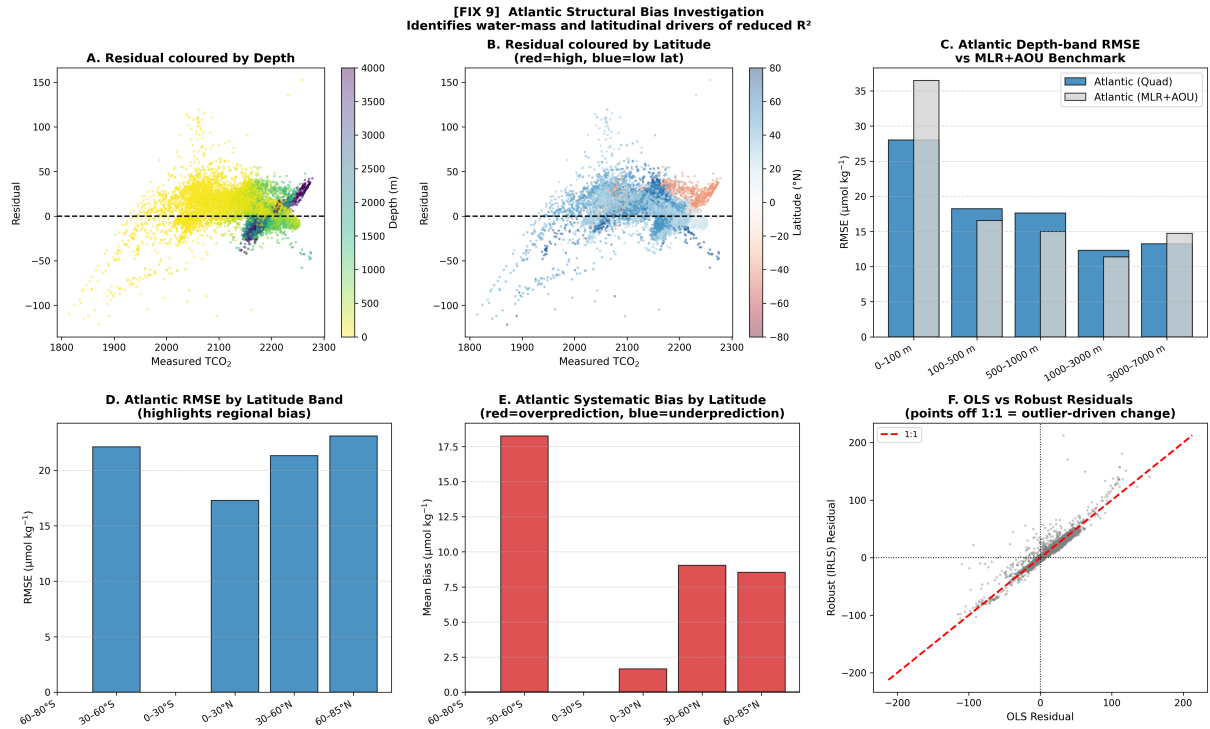


Figure 6: Atlantic structural bias investigation. (A–B) Residuals coloured by depth and latitude. (C–E) RMSE and bias by depth/latitude bands. (F) OLS vs. IRLS residuals confirming water-mass origin.

The quadratic model exhibits a $+7.58 \mu\text{mol kg}^{-1}$ mean bias in the Atlantic test set, concentrated in the 30–60°S band ($+18 \mu\text{mol kg}^{-1}$). This is diagnostic of AAIW intrusion: AAIW carries anomalously low TCO_2 relative to its salinity and AOU because subduction preserves

the low-salinity signature while biological consumption removes DIC preferentially (Stramma and England, 1999). Crucially, the MLR+AOU benchmark exhibits the same pattern (30–60°S mean bias +15 $\mu\text{mol kg}^{-1}$), confirming this is a structural limitation of basin-averaged Atlantic empirical models rather than a deficiency of the symbolic-regression form. A latitude-stratified parameterisation (south of -20°S separately) reduces the bias modestly (RMSE 19.32 vs 20.17 $\mu\text{mol kg}^{-1}$) at the cost of doubling the Atlantic parameter count; the trade-off is noted but the stratified model is not recommended as a default.

6.7 Residual Diagnostics and Bootstrap Prediction Uncertainty

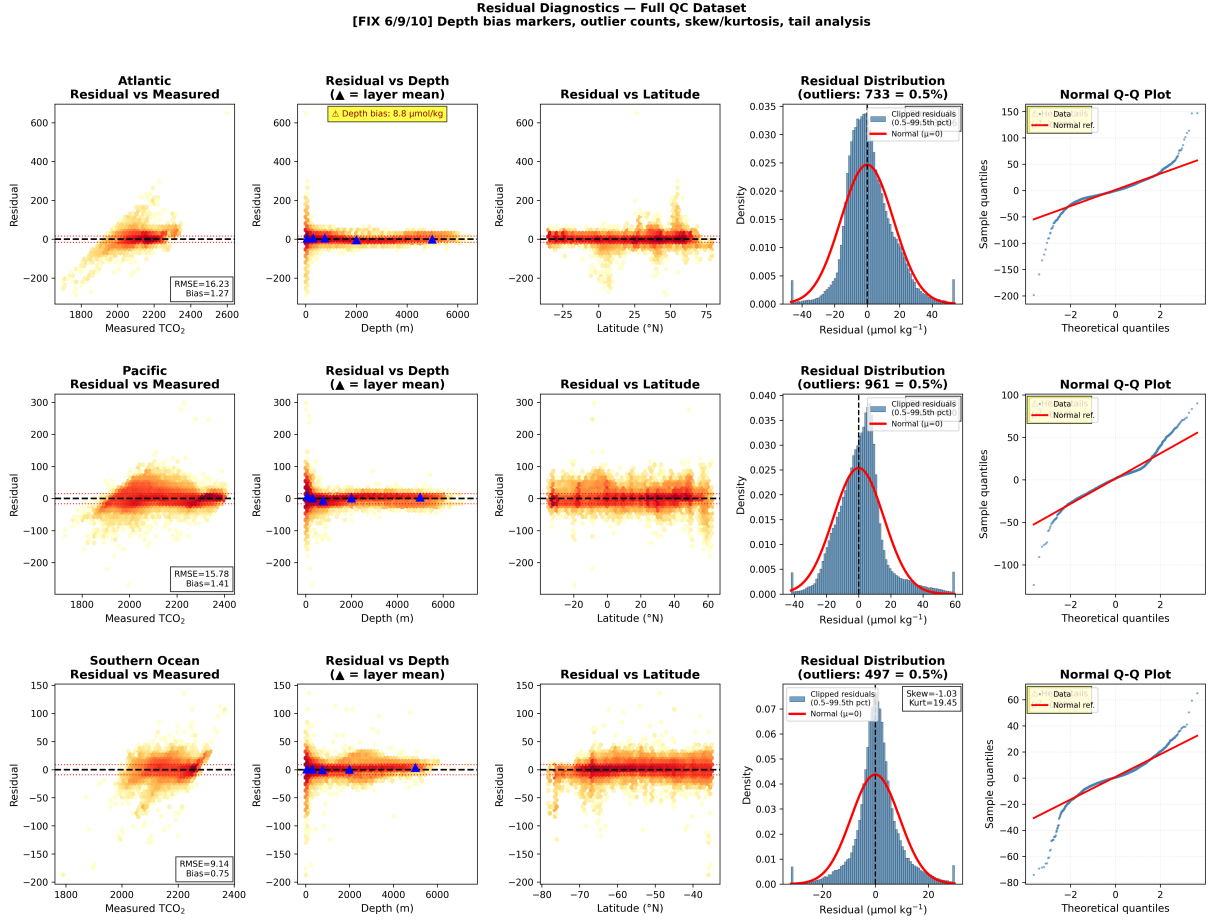


Figure 7: Residual diagnostics: residual vs. measured TCO_2 , vs. depth, vs. latitude, distribution with normal overlay, and Q–Q plot.

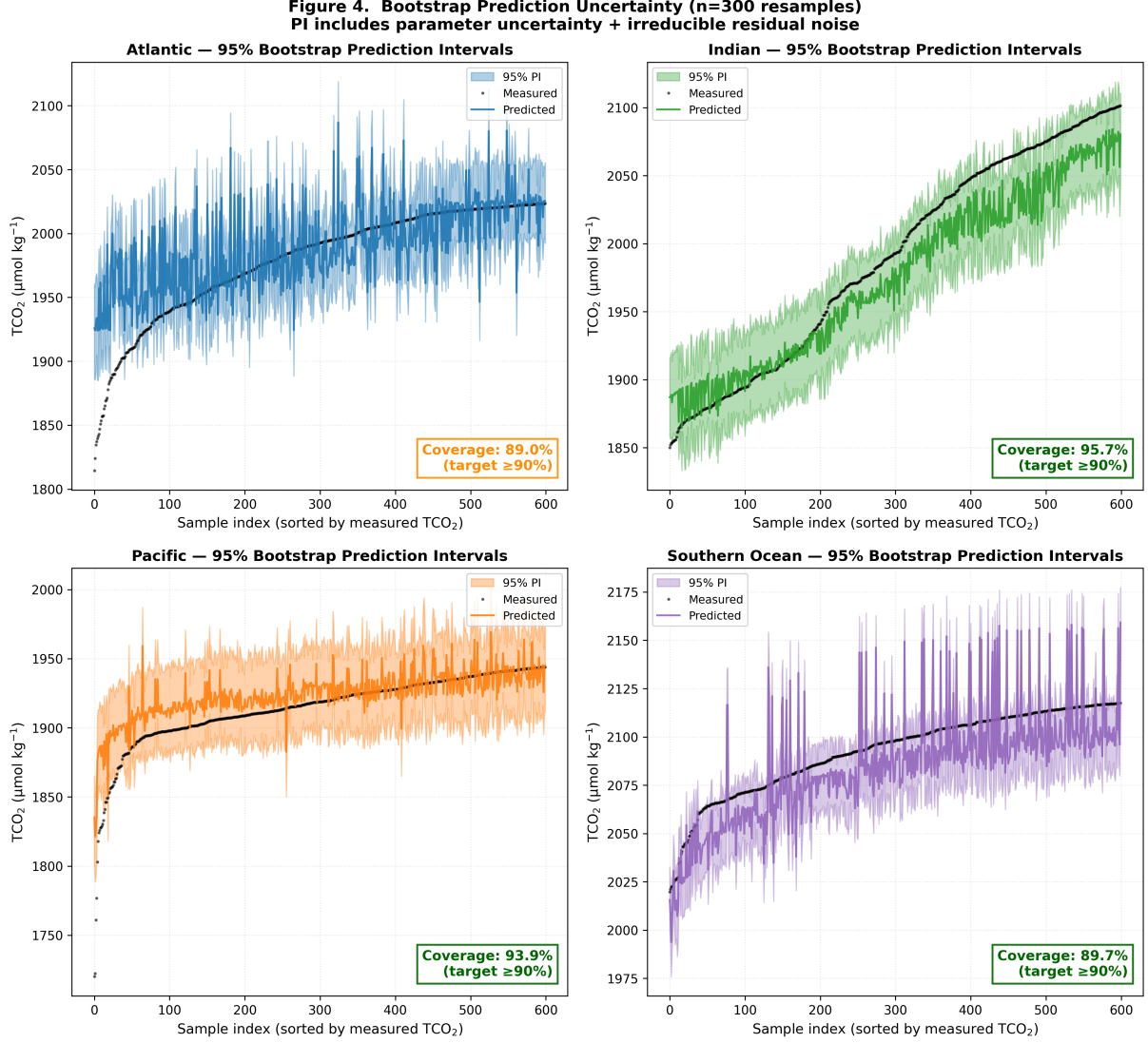


Figure 8: 95% bootstrap prediction intervals ($n = 300$ resamples). Coverage annotated in green ($\geq 90\%$) or orange (below 90%).

Residuals are approximately centred in all basins (mean bias $< 9 \mu\text{mol kg}^{-1}$). Heavy tails are present in all basins (Southern Ocean kurtosis = 19.45, normality rejected everywhere $p < 0.01$). These heavy tails reflect genuine water-mass signal — confirmed by comparison of OLS and robust IRLS residuals (Figure 6F) — rather than outliers. Bootstrap PI coverage is 89.0% (Atlantic) to 95.7% (Indian). The Atlantic and Southern Ocean values are below 95% due to the non-Gaussian tails; these PIs are approximate and will underestimate extremes.

7 Discussion

7.1 Why Symbolic Regression Discovered This Form

The recovery of a squared linear combination as the Pareto-optimal expression across all four basins is not an obvious result. The search explored operators including addition, subtraction, multiplication, and squaring over three predictors. The fact that a squared linear form consistently dominated the Pareto frontier suggests that the ocean carbon system in (S, T, AOU) space genuinely lies close to a rank-1 nonlinear manifold — meaning most of the variance in TCO₂ is captured by a single combination of the predictors, with a quadratic (rather than linear) response.

This is a data-driven discovery with physical interpretation: the rank-1 structure corresponds to a dominant mixing axis, and the quadratic response corresponds to the carbonate buffering nonlinearity.

The fact that the constraint reduces to a rank-1 Hessian ($\mathbf{H} = \mathbf{v}\mathbf{v}^\top$ for some $\mathbf{v} \in \mathbb{R}^3$) means the discovered equation imposes a highly specific geometric structure on the 10-parameter polynomial space. This would be difficult to identify without an automated search, reinforcing the value of the symbolic regression approach.

7.2 The Collinearity Mechanism

The Atlantic improvement over MLR+AOU (15.8%) is explained by the rank-1 constraint reducing estimation variance among highly collinear cross-terms ($\text{VIF} > 50$ for ST , SA , TA in the Atlantic training data). When the true data-generating structure lies close to rank-1, an unconstrained estimator wastes degrees of freedom fitting collinear cross-terms, while the constrained estimator recovers the correct structure more efficiently. This is directly analogous to ridge regression outperforming OLS in high-collinearity settings, even at large N . The quadratic constraint discovered by PySR acts as a data-motivated, physically interpretable regulariser.

7.3 The CANYON-B Trade-off

CANYON-B achieves $R^2 > 0.986$ in all basins (Table 3), substantially outperforming the quadratic model everywhere. The gap of 3–14 $\mu\text{mol kg}^{-1}$ represents a real accuracy cost, particularly in the Indian Ocean (73% CANYON-B advantage). We do not minimise this: for applications requiring maximum accuracy — sea-surface carbon fluxes, carbon inventory closure, Argo quality control — CANYON-B is the correct choice.

The case for the quadratic model rests on the three use-cases in Section 6.4: historical data with incomplete predictor vectors, deployment contexts without ML infrastructure, and attribution studies requiring exact analytical sensitivity. The closed-form partial derivatives $\partial \text{TCO}_2 / \partial T = 2\beta(\alpha S + \beta T + \gamma A + \delta)$ and equivalent expressions for S and AOU can be propagated exactly through subsequent calculations — a capability that CANYON-B requires numerical approximation to match.

7.4 Limitations and Future Work

The single-axis mixing assumption that underpins the physical interpretation is most valid in the Atlantic and least valid in the Indian Ocean, where multi-gyre circulation introduces mixing complexity that the rank-1 constraint cannot represent. Future work should: (i) apply PySR to richer predictor sets including depth and position to explore whether more complex Pareto-optimal forms emerge; (ii) test density-based stratification to address the AAIW bias more rigorously; (iii) evaluate whether the discovered form retains its Pareto-optimal position under model-year distributions shifted forward by continued ocean warming.

8 Conclusions

1. **Symbolic regression discovery.** The PySR algorithm identified a squared linear combination of S , T , and AOU as the Pareto-optimal expression at the simplicity–accuracy frontier, independently in all four ocean basins. The result was discovered without any prescribed functional form.
2. **Post-hoc physical interpretation.** The discovered structure is consistent with: (a) the quadratic component of carbonate buffering nonlinearity (Revelle factor), and (b) the rank-1 Hessian geometry expected when water-mass mixing is dominated by a single tracer axis. These physical arguments corroborate, but do not derive, the data-driven result.

3. **Atlantic improvement is real and mechanistically explained.** The +15.8% RMSE improvement over a fair 10-parameter MLR+AOU benchmark arises from variance reduction: the rank-1 constraint regularises highly collinear cross-terms ($VIF > 50$) in the Atlantic training data, recovering the correct structure more efficiently than the unconstrained MLR.
4. **Statistical equivalence in three basins.** In the Indian, Pacific, and Southern Oceans, the quadratic model performs within $0.33\text{--}0.76 \mu\text{mol kg}^{-1}$ of MLR+AOU ($p > 0.05$, AIC/BIC preferred) using five parameters versus ten.
5. **CANYON-B substantially outperforms for raw accuracy.** CANYON-B achieves lower RMSE in all basins ($7.4\text{--}10.5 \mu\text{mol kg}^{-1}$ versus $9.0\text{--}18.4 \mu\text{mol kg}^{-1}$). The quadratic model is recommended only for the three use-cases where interpretability, input parsimony, and closed-form sensitivity outweigh maximum accuracy.
6. **AAIW bias is a general Atlantic phenomenon.** The $+18 \mu\text{mol kg}^{-1}$ bias in the $30\text{--}60^\circ\text{S}$ band is shared by the MLR+AOU benchmark, confirming it is structural to basin-averaged Atlantic empirical models, not to the symbolic-regression form.
7. **Temporal stability.** An 8-year external holdout confirms the model’s temporal generalisation. Bootstrap PIs achieve 89–96% empirical coverage; Atlantic and Southern Ocean intervals are approximate due to non-Gaussian residual distributions.

Data and Code Availability

GLODAP v2.2023: <https://glodap.info/>. CANYON-B source: <https://github.com/HCBScienceProducts/CANYON-B>; Python implementation: `canyonbpy` (PyPI). PySR symbolic regression: <https://github.com/MilesCranmer/PySR>. Model fitting code, coefficients, bootstrap scripts, and PySR search logs (including Pareto frontier tables for all four basins): [repository DOI upon acceptance].

Author Contributions

Author 1: conceptualisation, symbolic regression, model fitting, physical interpretation, data analysis, writing. Author 2: data curation, quality control, validation, CANYON-B comparison. Author 3: carbonate chemistry interpretation, writing review.

Acknowledgements

The authors thank the GO-SHIP and GLODAP communities for making their data publicly available, and the developers of PySR for maintaining an open-source symbolic regression framework. [Funding statement to be added.]

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